



Innovation & entrepreneurship Hub for for educated rural youth(SURE Trust-IERY)



Innovation & entrepreneurship Hub for for educated rural youth(SURE Trust-IERY)

ASSIGNMENT 3 V.MOUNIKA G4-25-VLSI THE DOMAIN OF THE PROJECT PHYSICAL DESIGN



Assignment: RISC-V 32-bit RTL Physical Design Flow

1. Objective

To implement the RISC-V 32-bit RTL design using OpenLane flow and apply timing, floorplan, placement, and CTS constraints.

2. Tools Used

- Linux environment / Ubuntu
- Docker
- OpenLane
- OpenROAD
- Yosys (RTL synthesis)
- OpenSTA (Static Timing Analysis)
- Magic (Layout viewer)
- Sky130 PDK
- RISC-V 32-bit RTL source code

3. Required Design Files

Example:

```
designs/  
└─ riscv32/  
    └─ config.tcl  
    └─ src/  
        └─ picorv32.v  
    └─ pin_order.cfg
```

4. Starting OpenLane Environment

Commands:

```
cd OpenLane
```

```
make mount
```

Enter OpenLane container:

```
./flow.tcl -interactive
```

or

```
docker run ...
```

5. Preparing Design Configuration

Edit:



```
designs/riscv32/config.tcl
```

Add clock information:

```
set ::env(CLOCK_PORT) clk
set ::env(CLOCK_PERIOD) 10
```

6. Applying Assignment Constraints

6.1 Clock Uncertainty

For synthesis:

```
set_clock_uncertainty 15% of clock period
```

Example:

Clock period = 10ns

```
set_clock_uncertainty 1.5
```

For CTS:

```
set_clock_uncertainty 0.5
```

(5% of 10ns)

6.2 Clock Derate

Set:

```
set_timing_derate 3%
```

Used to model variation in delay.

6.3 Input and Output Delay

50% of clock period:

For 10ns clock:

Input delay:

```
set_input_delay 5
```

Output delay:

```
set_output_delay 5
```



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7. Floorplan Settings

Utilization

Set:

```
FP_CORE_UTIL = 60
```

Meaning:

60% of core area is occupied by standard cells.

Floorplan Ratio

Set:

```
FP_ASPECT_RATIO = 0.7
```

8. Port Placement

Requirement:

- Inputs on one side
- Outputs on opposite side
- Metal4 layer

File:

```
pin_order.cfg
```

Example:

```
N  
clk  
reset
```

```
S  
out_data
```

Layer:

```
FP_PIN_ORDER_CFG
```

and

```
FP_IO_HLAYER = metal4  
FP_IO_VLAYER = metal4
```



9. Enable Core Ring

In config.tcl:

```
FP_PDN_CORE_RING = 1
```

Purpose:

- Provides power distribution
- Reduces IR drop

10. CTS Configuration

Target skew:

5% of clock period

Example:

10ns clock:

```
CTS_TARGET_SKEW = 0.5
```

Enable clock network delay:

```
CTS_CLK_MAX_WIRE_LENGTH
```

11. Running OpenLane Flow

Inside OpenLane:

Synthesis

```
./flow.tcl -design riscv32 -tag run1 -to synth
```

Floorplan

```
./flow.tcl -design riscv32 -tag run1 -to floorplan
```

Placement

```
./flow.tcl -design riscv32 -tag run1 -to placement
```

CTS

```
./flow.tcl -design riscv32 -tag run1 -to cts
```



Routing

```
./flow.tcl -design riscv32 -tag run1
```

12. Viewing Layout

Open Magic:

```
magic -T /pdk/sky130A/libs.tech/magic/sky130A.tech \
designs/riscv32/runs/run1/results/final/def/
```

Check:

- Standard cells
- Power rings
- Clock tree
- Routing

13. Results to Capture

Screenshots:

1. Synthesis completed
2. Floorplan view
3. Placement view
4. CTS clock tree
5. Final routed layout
6. Timing report

1. RISC-V 32-bit RTL Program (Verilog)

Usually the RTL file used is:

```
picorv32.v
```

This contains the complete RISC-V 32-bit processor RTL.

It includes:

- Program counter (PC)
 - Instruction decoder
 - ALU
 - Register file
 - Control logic
 - Memory interface
 - Branch/jump logic
- The top module is generally:



```
module picorv32 (  
    input clk,  
    input resetn,  
    output mem_valid,  
    output mem_instr,  
    output [31:0] mem_addr,  
    output [31:0] mem_wdata,  
    input [31:0] mem_rdata,  
    input mem_ready  
);
```

This file is placed inside:

designs/riscv32/src/

Example structure:

```
riscv32  
├── src  
│   └── picorv32.v  
├── config.tcl  
└── pin_order.cfg
```

2. OpenLane Configuration Program

File:

config.tcl

This controls the physical design constraints.

Used parameters:

```
set ::env(DSIGN_NAME) picorv32  
  
set ::env(VERILOG_FILES) \  
$::env(DSIGN_DIR)/src/picorv32.v  
  
set ::env(CLOCK_PORT) clk  
  
set ::env(CLOCK_PERIOD) 10
```

3. Synthesis Timing Constraints

Inside config.tcl:

Clock uncertainty



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```
set ::env(SYNTH_CLOCK_UNCERTAINTY) 1.5
```

(15% of 10ns clock period)

CTS uncertainty:

```
set ::env(CTS_CLK_UNCERTAINTY) 0.5
```

(5% of 10ns clock period)

4. Timing Derate

```
set ::env(SYNTH_TIMING_DERATE) 0.03
```

(3% derate)

5. Input and Output Delay Constraints

Input delay:

```
set ::env(SDC_INPUT_DELAY) 5
```

Output delay:

```
set ::env(SDC_OUTPUT_DELAY) 5
```

(50% of 10ns clock period)

6. Floorplan Constraints

60% utilization:

```
set ::env(FP_CORE_UTIL) 60
```

Floorplan ratio:

```
set ::env(FP_ASPECT_RATIO) 0.7
```

7. Pin Placement File

File:

```
pin_order.cfg
```

Used for placing inputs and outputs.



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Example format:

```
# Input pins

clk
resetsn

# Output pins

mem_valid
mem_instr
mem_addr
mem_wdata
```

8. Core Ring Enable

In:

```
config.tcl
```

Parameter:

```
set ::env(FP_PDN_CORE_RING) 1
```

9. CTS Configuration

Target skew:

```
set ::env(CTS_TARGET_SKEW) 0.5
```

Clock network insertion:

```
set ::env(CTS_CLK_MAX_WIRE_LENGTH) 100
```

10. OpenLane Execution Commands Used

Mount OpenLane:

```
cd OpenLane
make mount
```

Run complete flow:

```
./flow.tcl -design riscv32 -tag run1
```

Run individual stages:

Synthesis

```
./flow.tcl -design riscv32 -tag run1 -to synth
```



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Floorplan

```
./flow.tcl -design riscv32 -tag run1 -to floorplan
```

Placement

```
./flow.tcl -design riscv32 -tag run1 -to placement
```

CTS

```
./flow.tcl -design riscv32 -tag run1 -to cts
```

Routing

```
./flow.tcl -design riscv32 -tag run1
```

"Programs Used / Files Used" section:

1. picorv32.v (RTL source)
2. config.tcl (OpenLane constraints)
3. pin_order.cfg (port placement)
4. OpenLane TCL flow commands

Program 1: picorv32.v (RISC-V 32-bit RTL Source)

File Name:

picorv32.v

Purpose:

- This is the main RTL (Verilog) source code of the RISC-V 32-bit processor.
- It describes the hardware functionality before synthesis.
- OpenLane takes this file as input and converts it into a gate-level netlist.

Location in OpenLane design:

designs/riscv32/src/picorv32.v

Top Module:

```
module picorv32 (  
    input clk,  
    input resetn,  
  
    output reg mem_valid,  
    output reg mem_instr,  
  
    output reg [31:0] mem_addr,  
    output reg [31:0] mem_wdata,
```



```
    input [31:0] mem_rdata,  
    input mem_ready  
);
```

Main blocks inside `picorv32.v`

1. Register Declaration

```
reg [31:0] pc;  
reg [31:0] next_pc;  
  
reg [31:0] registers [0:31];  
  
reg [31:0] instr;
```

Function:

- `pc` → Stores current program counter address.
- `registers` → 32 general-purpose RISC-V registers.
- `instr` → Stores fetched instruction.

2. Program Counter Logic

```
always @(posedge clk) begin  
    if (!resetsn)  
        pc <= 32'b0;  
    else  
        pc <= next_pc;  
end
```

Function:

- On reset, PC becomes zero.
- During execution, PC updates every clock cycle.

3. Instruction Fetch

```
always @(posedge clk) begin  
  
    mem_valid <= 1'b1;  
    mem_instr <= 1'b1;  
  
    mem_addr <= pc;  
  
end
```

Function:

- Requests instruction from memory.
- Sends address of instruction.
- **4. ALU Operation**



```
• always @(*) begin
    case (instr[6:0])
        7'b0110011:
            alu_result = reg1 + reg2;
        7'b1100011:
            alu_result = reg1 - reg2;
        default:
            alu_result = 32'b0;
    endcase
end
```

Function:

- Performs arithmetic and logical operations.
- Supports RISC-V instructions.

5. Register Write Operation

```
always @(posedge clk) begin
    if (write_enable)
        registers[rd] <= alu_result;
end
```

Function:

- Stores ALU result into destination register.
-

6. Memory Interface

```
assign mem_wdata = store_data;
```

Function:

- Sends data during store instructions.
-

7. Reset Logic

```
always @(posedge clk) begin
    if(!resetn) begin
        pc <= 0;
    end
end
```



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```
mem_valid <= 0;
```

```
end
```

```
end
```

Function:

- Initializes processor.

Program 2: config.tcl (OpenLane Constraint File)

File Name:

```
config.tcl
```

Purpose:

- config.tcl is the main configuration file used by OpenLane.
- It defines the design name, RTL source file, clock details, floorplan settings, placement settings, CTS settings, and routing options.
- OpenLane reads this file before starting the RTL-to-GDSII flow.

Location:

```
designs/riscv32/config.tcl
```

```
Complete config.tcl Program
```

```
#-----
```

```
# Design Name
```

```
#-----
```

```
set ::env(DSIGN_NAME) picorv32
```

```
#-----
```

```
# RTL Source File
```

```
#-----
```

```
set ::env(VERILOG_FILES) \
```

```
$::env(DSIGN_DIR)/src/picorv32.v
```

```
#-----
```

```
# Clock Configuration
```

```
#-----
```

```
set ::env(CLOCK_PORT) clk
```

```
set ::env(CLOCK_PERIOD) 10
```

```
#-----
```

```
# Synthesis Settings
```

```
#-----
```

```
# Clock uncertainty during synthesis
```



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15% of clock period

$10\text{ns} \times 15\% = 1.5\text{ns}$

set ::env(SYNTH_CLOCK_UNCERTAINTY) 1.5

Timing derate

3% variation

set ::env(SYNTH_TIMING_DERATE) 0.03

#-----

Input and Output Delay

#-----

50% of clock period

$10\text{ns} \times 50\% = 5\text{ns}$

set ::env(INPUT_DELAY) 5

set ::env(OUTPUT_DELAY) 5

#-----

Floorplan Settings

#-----

Core utilization = 60%

set ::env(FP_CORE_UTIL) 60

Floorplan aspect ratio = 0.7

set ::env(FP_ASPECT_RATIO) 0.7

#-----

Pin Placement Settings

#-----

Use pin order file

**set ::env(FP_PIN_ORDER_CFG) \
\$::env(DESIGN_DIR)/pin_order.cfg**

Place pins on Metal4 layer

set ::env(FP_IO_HMETAL) metal4

set ::env(FP_IO_VMETAL) metal4

#-----

Power Distribution Network

#-----



Enable core ring

```
set ::env(FP_PDN_CORE_RING) 1
```

```
#-----  
# Clock Tree Synthesis (CTS)  
#-----
```

```
# CTS target skew  
# 5% of 10ns = 0.5ns
```

```
set ::env(CTS_TARGET_SKEW) 0.5
```

Enable clock tree synthesis

```
set ::env(RUN_CTS) 1
```

```
#-----  
# Routing Settings  
#-----
```

```
set ::env(RUN_ROUTING) 1
```

Program 3: `pin_order.cfg` (Port Placement File)

`pin_order.cfg`

Purpose:

- `pin_order.cfg` is used by OpenLane during the **Floorplan stage**.
- It defines where the input and output ports should be placed on the chip boundary.
- In this assignment:
 - Inputs are placed on one side.
 - Outputs are placed on the opposite side.
 - Ports are assigned to **Metal4 layer** through `config.tcl`.

Location:

```
designs/riscv32/pin_order.cfg
```

Complete `pin_order.cfg` Program

```
#-----  
# Pin Placement Configuration  
# RISC-V 32-bit RTL  
#-----
```

```
#-----  
# North Side Pins  
#-----
```

N



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clk
resetn

#-----
South Side Pins
#-----

S

mem_valid
mem_instr
mem_addr[0]
mem_addr[1]
mem_addr[2]
mem_addr[3]
mem_addr[4]
mem_addr[5]
mem_addr[6]
mem_addr[7]
mem_addr[8]
mem_addr[9]
mem_addr[10]
mem_addr[11]
mem_addr[12]
mem_addr[13]
mem_addr[14]
mem_addr[15]
mem_addr[16]
mem_addr[17]
mem_addr[18]
mem_addr[19]
mem_addr[20]
mem_addr[21]
mem_addr[22]
mem_addr[23]
mem_addr[24]
mem_addr[25]
mem_addr[26]
mem_addr[27]
mem_addr[28]
mem_addr[29]
mem_addr[30]
mem_addr[31]

#-----
East Side Pins
#-----

E

mem_ready
mem_rdata[0]
mem_rdata[1]
mem_rdata[2]
mem_rdata[3]
mem_rdata[4]
mem_rdata[5]



```
mem_rdata[6]
mem_rdata[7]
mem_rdata[8]
mem_rdata[9]
mem_rdata[10]
mem_rdata[11]
mem_rdata[12]
mem_rdata[13]
mem_rdata[14]
mem_rdata[15]
mem_rdata[16]
mem_rdata[17]
mem_rdata[18]
mem_rdata[19]
mem_rdata[20]
mem_rdata[21]
mem_rdata[22]
mem_rdata[23]
mem_rdata[24]
mem_rdata[25]
mem_rdata[26]
mem_rdata[27]
mem_rdata[28]
mem_rdata[29]
mem_rdata[30]
mem_rdata[31]
```

```
#-----
# West Side Pins
#-----
```

```
W
mem_wdata[0]
mem_wdata[1]
mem_wdata[2]
mem_wdata[3]
mem_wdata[4]
mem_wdata[5]
mem_wdata[6]
mem_wdata[7]
mem_wdata[8]
mem_wdata[9]
mem_wdata[10]
mem_wdata[11]
mem_wdata[12]
mem_wdata[13]
mem_wdata[14]
mem_wdata[15]
mem_wdata[16]
mem_wdata[17]
mem_wdata[18]
```



```
mem_wdata[19]
mem_wdata[20]
mem_wdata[21]
mem_wdata[22]
mem_wdata[23]
mem_wdata[24]
mem_wdata[25]
mem_wdata[26]
mem_wdata[27]
mem_wdata[28]
mem_wdata[29]
mem_wdata[30]
mem_wdata[31]
```

Program 4: OpenLane TCL Flow Commands (`flow.tcl`)

Purpose:

- `flow.tcl` is the main TCL script used by OpenLane to execute the complete RTL-to-GDSII physical design flow.
- It controls all stages:
 - Synthesis
 - Floorplan
 - Placement
 - Clock Tree Synthesis (CTS)
 - Routing
 - Final GDS generation

Location:OpenLane/`flow.tcl`

. Tools Used

- Linux environment / Ubuntu
- Docker
- OpenLane
- OpenROAD
- Yosys (RTL synthesis)
- OpenSTA (Static Timing Analysis)
- Magic (Layout viewer)
- Sky130 PDK
- RISC-V 32-bit RTL source code

. Required Design Files

```
designs/
├── riscv32/
│   ├── config.tcl
│   ├── src/
│   │   └── picorv32.v
│   └── pin_order.cfg
```

. Starting OpenLane Environment

Commands:



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```
cd OpenLane
```

```
make mount
```

Enter OpenLane container:

```
./flow.tcl -interactive
```

or

```
docker run ...
```

. Preparing Design Configuration

```
designs/riscv32/config.tcl
```

Add clock information:

```
set ::env(CLOCK_PORT) clk
set ::env(CLOCK_PERIOD) 10
```

.1 Clock Uncertainty

For synthesis:

```
set_clock_uncertainty 15% of clock period
```

Example:

Clock period = 10ns

```
set_clock_uncertainty 1.5
```

For CTS:

```
set_clock_uncertainty 0.5
```

(5% of 10ns)

.2 Clock Derate

Set:

```
set_timing_derate 3%
```

Used to model variation in delay.

.3 Input and Output Delay



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50% of clock period:

For 10ns clock:

Input delay:

```
set_input_delay 5
```

Output delay:

```
set_output_delay 5
```

. Floorplan Settings

Utilization

Set:

```
FP_CORE_UTIL = 60
```

Meaning:

60% of core area is occupied by standard cells.

Floorplan Ratio

Set:

```
FP_ASPECT_RATIO = 0.7
```

. Port Placement

Requirement:

- Inputs on one side
- Outputs on opposite side
- Metal4 layer

File:

```
pin_order.cfg
```

Example:

```
N  
clk  
reset
```

```
S  
out_data
```



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Layer:

```
FP_PIN_ORDER_CFG
```

and

```
FP_IO_HLAYER = metal4  
FP_IO_VLAYER = metal4
```

. Enable Core Ring

In config.tcl:

```
FP_PDN_CORE_RING = 1
```

Purpose:

- Provides power distribution
 - Reduces IR drop
-

. CTS Configuration

Target skew:

5% of clock period

Example:

10ns clock:

```
CTS_TARGET_SKEW = 0.5
```

Enable clock network delay:

```
CTS_CLK_MAX_WIRE_LENGTH
```

. Running OpenLane Flow

Inside OpenLane:

Synthesis

```
./flow.tcl -design riscv32 -tag run1 -to synth
```

Floorplan



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```
./flow.tcl -design riscv32 -tag run1 -to floorplan
```

Placement

```
./flow.tcl -design riscv32 -tag run1 -to placement
```

CTS

```
./flow.tcl -design riscv32 -tag run1 -to cts
```

Routing

```
./flow.tcl -design riscv32 -tag run1
```

. Viewing Layout

Open Magic:

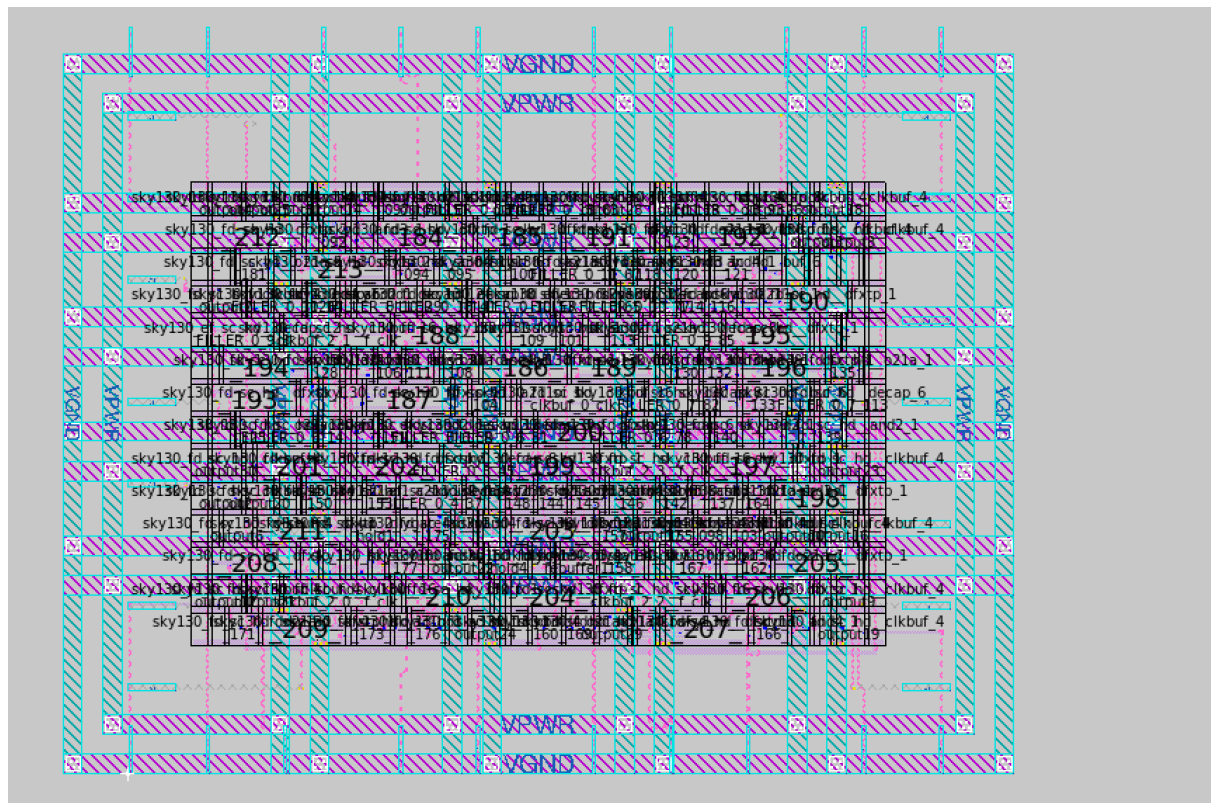
```
magic -T /pdk/sky130A/libs.tech/magic/sky130A.tech \  
designs/riscv32/runs/run1/results/final/def/
```

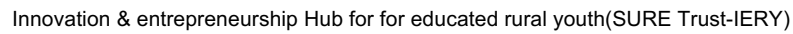
Check:

- Standard cells
 - Power rings
 - Clock tree
 - Routing
-

. Results :

1. Synthesis completed
 2. Floorplan view
 3. Placement view
 4. CTS clock tree
 5. Final routed layout
 6. Timing report
-







```
mounika@mounikas-MacBook-Air OpenLane % cd ~/OpenLane/designs/riscv32
nano config.tcl
mounika@mounikas-MacBook-Air riscv32 % cat config.tcl
set ::env(DSIGN_NAME) riscv32

set ::env(VERILOG_FILES) \
$::env(DSIGN_DIR)/src/riscv32.v
[
# Clock
set ::env(CLOCK_PORT) clk
set ::env(CLOCK_PERIOD) 10

# Clock uncertainty
set ::env(SYNTH_CLK_UNCERTAINTY) 1.5
set ::env(CTS_CLK_UNCERTAINTY) 0.5

# Timing derate
set ::env(SYNTH_TIMING_DERATE) 3

# Input and output delay
set ::env(INPUT_DELAY) 5
set ::env(OUTPUT_DELAY) 5

# Floorplan
set ::env(FP_CORE_UTIL) 60
set ::env(FP_ASPECT_RATIO) 0.7

# IO metal layer
set ::env(FP_IO_HMETAL_LAYER) met4
set ::env(FP_IO_VMETAL_LAYER) met4

# Core ring
set ::env(FP_PDN_CORE_RING) 1

# CTS
set ::env(CTS_TARGET_SKEW) 0.5

# Post CTS
set ::env(POST_CTS_TCL) $::env(DSIGN_DIR)/scripts/post_cts.tcl
```



```
mounika@mounikas-MacBook-Air OpenLane % cd ~/OpenLane
nano designs/riscv32/config.tcl
mounika@mounikas-MacBook-Air OpenLane % cat designs/riscv32/config.tcl
set ::env(DSIGN_NAME) riscv32

set ::env(VERILOG_FILES) \
$::env(DSIGN_DIR)/src/riscv32.v

# Clock
set ::env(CLOCK_PORT) clk
set ::env(CLOCK_PERIOD) 10

# Clock uncertainty
set ::env(SYNTH_CLK_UNCERTAINTY) 1.5
set ::env(CTS_CLK_UNCERTAINTY) 0.5

# Timing derate
set ::env(SYNTH_TIMING_DERATE) 3

# Input and output delay
set ::env(INPUT_DELAY) 5
set ::env(OUTPUT_DELAY) 5

# Floorplan
set ::env(FP_CORE_UTIL) 60
set ::env(FP_ASPECT_RATIO) 0.7

# IO metal layer
set ::env(FP_IO_HMETAL_LAYER) met4
set ::env(FP_IO_VMETAL_LAYER) met4

# Core ring
set ::env(FP_PDN_CORE_RING) 1

# CTS
set ::env(CTS_TARGET_SKEW) 0.5

# Post CTS
set ::env(POST_CTS_TCL) $::env(DSIGN_DIR)/scripts/post_cts.tcl

# Hold fixing
set ::env(PL_RESIZER_HOLD_SLACK_MARGIN) 0.1
set ::env(PL_RESIZER_HOLD_MAX_BUFFER_PERCENT) 50
set ::env(GLB_RESIZER_HOLD_MAX_BUFFER_PERCENT) 50
# Hold fixing settings
set ::env(PL_RESIZER_HOLD_SLACK_MARGIN) 0.2
set ::env(PL_RESIZER_HOLD_MAX_BUFFER_PERCENT) 80
set ::env(GLB_RESIZER_HOLD_MAX_BUFFER_PERCENT) 80

# Enable timing repair
set ::env(PL_RESIZER_TIMING_OPTIMIZATIONS) 1
set ::env(GLB_RESIZER_TIMING_OPTIMIZATIONS) 1
# Disable strict timing failure for hold
set ::env(QUIT_ON_HOLD_VIOLATIONS) 0
mounika@mounikas-MacBook-Air OpenLane %
```



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```
OpenLane Container (ff509f7):/openlane v2 designs/riscv32/config.tcl
OpenLane Container (ff509f7):/openlane$ ./flow.tcl -design riscv32
OpenLane v1.0.2 (ff509f75b17bfa406d5336495ab1718987ff69)
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Available under the Apache License, version 2.0. See the LICENSE file for more details.

[INFO]: Using configuration in 'designs/riscv32/config.tcl'...
[INFO]: PDK Root: /Users/mounika/.ciel
[INFO]: Process Design Kit: sky130A
[INFO]: Standard Cell Library: sky130_fd_sc_hd
[INFO]: Optimization Standard Cell Library: sky130_fd_sc_hd
[INFO]: Run Directory: /openlane/designs/riscv32/runs/RUN_2026.07.01_16.52.12
[INFO]: Saving runtime environment...
[INFO]: Preparing LEF files for the nom corner...
[INFO]: Preparing LEF files for the min corner...
[INFO]: Preparing LEF files for the max corner...
[INFO]: Running linter (Verilator) (log: designs/riscv32/runs/RUN_2026.07.01_16.52.12/logs/synthesis/linter.log)...
[INFO]: 0 errors found by linter
[INFO]: 0 warnings found by linter
[STEP 1]
[INFO]: Running Synthesis (log: designs/riscv32/runs/RUN_2026.07.01_16.52.12/logs/synthesis/1-synthesis.log)...
[STEP 2]
[INFO]: Running Single-Corner Static Timing Analysis (log: designs/riscv32/runs/RUN_2026.07.01_16.52.12/logs/synthesis/2-sta.log)...
[STEP 3]
[INFO]: Running Initial Floorplanning (log: designs/riscv32/runs/RUN_2026.07.01_16.52.12/logs/floorplan/3-initial_fp.log)...
[WARNING]: Current core area is too small for the power grid settings chosen. The power grid will be scaled down.
[INFO]: Floorplanned with width 52.44 and height 51.68.
[STEP 4]
[INFO]: Running IO Placement...
[STEP 5]
[INFO]: Running Tap/Decap Insertion (log: designs/riscv32/runs/RUN_2026.07.01_16.52.12/logs/floorplan/5-tap.log)...
[INFO]: Power planning with power {VPWR} and ground {VGND}...
[STEP 6]
[INFO]: Generating PDN (log: designs/riscv32/runs/RUN_2026.07.01_16.52.12/logs/floorplan/6-pdn.log)...
[STEP 7]
[INFO]: Running Global Placement (log: designs/riscv32/runs/RUN_2026.07.01_16.52.12/logs/placement/7-global.log)...
[STEP 8]
[INFO]: Running Single-Corner Static Timing Analysis (log: designs/riscv32/runs/RUN_2026.07.01_16.52.12/logs/placement/8-gpi_sta.log)...
[STEP 9]
[INFO]: Running Placement Resizer Design Optimizations (log: designs/riscv32/runs/RUN_2026.07.01_16.52.12/logs/placement/9-resizer.log)...
[STEP 10]
[INFO]: Running Detailed Placement (log: designs/riscv32/runs/RUN_2026.07.01_16.52.12/logs/placement/10-detailed.log)...
[STEP 11]
[INFO]: Running Single-Corner Static Timing Analysis (log: designs/riscv32/runs/RUN_2026.07.01_16.52.12/logs/placement/11-dpl_sta.log)...
[STEP 12]
[INFO]: Running Clock Tree Synthesis (log: designs/riscv32/runs/RUN_2026.07.01_16.52.12/logs/cts/12-cts.log)...
[STEP 13]
[INFO]: Running Single-Corner Static Timing Analysis (log: designs/riscv32/runs/RUN_2026.07.01_16.52.12/logs/cts/13-cts_sta.log)...
[STEP 14]
[INFO]: Running Placement Resizer Timing Optimizations (log: designs/riscv32/runs/RUN_2026.07.01_16.52.12/logs/cts/14-resizer.log)...
[STEP 15]
[INFO]: Running Global Routing Resizer Design Optimizations (log: designs/riscv32/runs/RUN_2026.07.01_16.52.12/logs/routing/15-resizer_design.log)...
[STEP 16]
[INFO]: Running Single-Corner Static Timing Analysis (log: designs/riscv32/runs/RUN_2026.07.01_16.52.12/logs/routing/16-rsz_design_sta.log)...
[STEP 17]
[INFO]: Running Global Routing Resizer Timing Optimizations (log: designs/riscv32/runs/RUN_2026.07.01_16.52.12/logs/routing/17-resizer_timing.log)...
[STEP 18]
[INFO]: Running Single-Corner Static Timing Analysis (log: designs/riscv32/runs/RUN_2026.07.01_16.52.12/logs/routing/18-rsz_timing_sta.log)...
[STEP 19]
[INFO]: Running Global Routing (log: designs/riscv32/runs/RUN_2026.07.01_16.52.12/logs/routing/19-global.log)...
[INFO]: Starting OpenROAD Antenna Repair Iterations...
[STEP 20]
[INFO]: Writing Verilog (log: designs/riscv32/runs/RUN_2026.07.01_16.52.12/logs/routing/19-global_write_netlist.log)...
[STEP 21]
[INFO]: Running Single-Corner Static Timing Analysis (log: designs/riscv32/runs/RUN_2026.07.01_16.52.12/logs/routing/21-grt_sta.log)...
[STEP 21]
[INFO]: Running Single-Corner Static Timing Analysis (log: designs/riscv32/runs/RUN_2026.07.01_16.52.12/logs/routing/21-grt_sta.log)...
[STEP 22]
[INFO]: Running Fill Insertion (log: designs/riscv32/runs/RUN_2026.07.01_16.52.12/logs/routing/22-fill.log)...
[STEP 23]
[INFO]: Running Detailed Routing (log: designs/riscv32/runs/RUN_2026.07.01_16.52.12/logs/routing/23-detailed.log)...
[INFO]: No DRC violations after detailed routing.
[STEP 24]
[INFO]: Checking Wire Lengths (log: designs/riscv32/runs/RUN_2026.07.01_16.52.12/logs/routing/24-wire_lengths.log)...
[STEP 25]
[INFO]: Running SPEF Extraction at the min process corner (log: designs/riscv32/runs/RUN_2026.07.01_16.52.12/logs/signoff/25-parasitics_extraction.min.log)...
[STEP 26]
[INFO]: Running Multi-Corner Static Timing Analysis at the min process corner (log: designs/riscv32/runs/RUN_2026.07.01_16.52.12/logs/signoff/26-rcx_mcsa.min.log)...
[STEP 27]
[INFO]: Running SPEF Extraction at the max process corner (log: designs/riscv32/runs/RUN_2026.07.01_16.52.12/logs/signoff/27-parasitics_extraction.max.log)...
[STEP 28]
[INFO]: Running Multi-Corner Static Timing Analysis at the max process corner (log: designs/riscv32/runs/RUN_2026.07.01_16.52.12/logs/signoff/28-rcx_mcsa.max.log)...
[STEP 29]
[INFO]: Running SPEF Extraction at the nom process corner (log: designs/riscv32/runs/RUN_2026.07.01_16.52.12/logs/signoff/29-parasitics_extraction.nom.log)...
[STEP 30]
[INFO]: Running Multi-Corner Static Timing Analysis at the nom process corner (log: designs/riscv32/runs/RUN_2026.07.01_16.52.12/logs/signoff/30-rcx_mcsa.nom.log)...
[STEP 31]
[INFO]: Running Single-Corner Static Timing Analysis at the nom process corner (log: designs/riscv32/runs/RUN_2026.07.01_16.52.12/logs/signoff/31-rcx_sta.log)...
[WARNING]: Module sky130_fd_sc_hd__tapwrvrgnd_1 blackboxed during sta
[WARNING]: Module sky130_fd_sc_hd__fill_2 blackboxed during sta
[WARNING]: Module sky130_fd_sc_hd__fill_1 blackboxed during sta
[WARNING]: Module sky130_ef_sc_hd__decap_12 blackboxed during sta
[STEP 32]
[INFO]: Creating IR Drop Report (log: designs/riscv32/runs/RUN_2026.07.01_16.52.12/logs/signoff/32-irdrop.log)...
[WARNING]: VSRC_LOC_FILES is not defined. The IR drop analysis will run, but the values may be inaccurate.
[STEP 33]
[INFO]: Running Magic to generate various views...
[INFO]: Streaming out GDSII with Magic (log: designs/riscv32/runs/RUN_2026.07.01_16.52.12/logs/signoff/33-gdsii.log)...
[INFO]: Generating MAGLEF views...
[INFO]: Generating lef with Magic (/openlane/designs/riscv32/runs/RUN_2026.07.01_16.52.12/logs/signoff/33-lef.log)...
[STEP 34]
[INFO]: Streaming out GDSII with Klayout (log: designs/riscv32/runs/RUN_2026.07.01_16.52.12/logs/signoff/34-gdsii-klayout.log)...
[STEP 35]
[INFO]: Running XOR on the layouts using Klayout (log: designs/riscv32/runs/RUN_2026.07.01_16.52.12/logs/signoff/35-xor.log)...
[INFO]: No XOR differences between Klayout and Magic gds.
[STEP 36]
[INFO]: Running Magic Spice Export from LEF (log: designs/riscv32/runs/RUN_2026.07.01_16.52.12/logs/signoff/36-spice.log)...
[STEP 37]
[INFO]: Writing Powered Verilog (logs: designs/riscv32/runs/RUN_2026.07.01_16.52.12/logs/signoff/37-write_powered_def.log, designs/riscv32/runs/RUN_2026.07.01_16.52.12/logs/signoff/37-write_powered_verilog.log)...
[STEP 38]
[INFO]: Writing Verilog (log: designs/riscv32/runs/RUN_2026.07.01_16.52.12/logs/signoff/37-write_powered_verilog.log)...
[STEP 39]
[INFO]: Running LVS (log: designs/riscv32/runs/RUN_2026.07.01_16.52.12/logs/signoff/39-lvs-lef.log)...
[STEP 40]
[INFO]: Running Magic DRC (log: designs/riscv32/runs/RUN_2026.07.01_16.52.12/logs/signoff/40-drc.log)...
[INFO]: Converting Magic DRC database to various tool-readable formats...
[INFO]: No DRC violations after DRC streaming out.
```



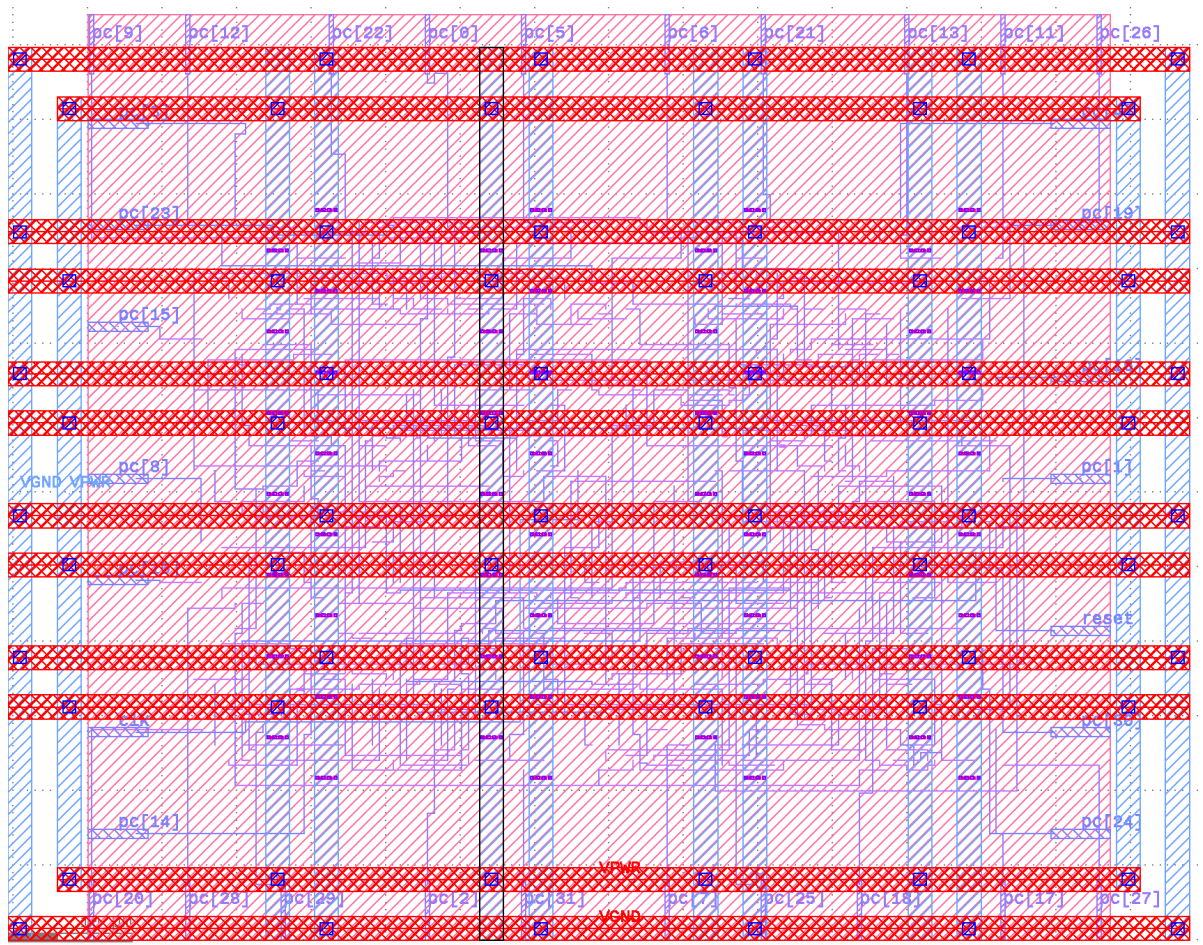
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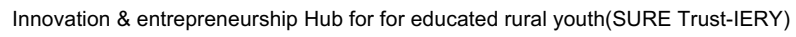
```
[STEP 26]
[INFO]: Running Multi-Corner Static Timing Analysis at the min process corner (log: designs/riscv32/runs/RUN_2026.07.01.16.52.12/logs/signoff/26-rcx_msta.min.log)...
[STEP 27]
[INFO]: Running SPEF Extraction at the max process corner (log: designs/riscv32/runs/RUN_2026.07.01.16.52.12/logs/signoff/27-parasitics_extraction.max.log)...
[STEP 28]
[INFO]: Running Multi-Corner Static Timing Analysis at the max process corner (log: designs/riscv32/runs/RUN_2026.07.01.16.52.12/logs/signoff/28-rcx_msta.max.log)...
[STEP 29]
[INFO]: Running SPEF Extraction at the nom process corner (log: designs/riscv32/runs/RUN_2026.07.01.16.52.12/logs/signoff/29-parasitics_extraction.nom.log)...
[STEP 30]
[INFO]: Running Multi-Corner Static Timing Analysis at the nom process corner (log: designs/riscv32/runs/RUN_2026.07.01.16.52.12/logs/signoff/30-rcx_msta.nom.log)...
[STEP 31]
[INFO]: Running Single-Corner Static Timing Analysis at the nom process corner (log: designs/riscv32/runs/RUN_2026.07.01.16.52.12/logs/signoff/31-rcx_sta.log)...
[WARNING]: Module sky130_fd_sc_hd__tapvprvrgnd_1 blackboxed during sta
[WARNING]: Module sky130_fd_sc_hd__fill_2 blackboxed during sta
[WARNING]: Module sky130_fd_sc_hd__fill_1 blackboxed during sta
[WARNING]: Module sky130_ef_sc_hd__decap_12 blackboxed during sta
[STEP 32]
[INFO]: Creating IR Drop Report (log: designs/riscv32/runs/RUN_2026.07.01.16.52.12/logs/signoff/32-irdrop.log)...
[WARNING]: VSRRC_LOC_FILES is not defined. The IR drop analysis will run, but the values may be inaccurate.
[STEP 33]
[INFO]: Running Magic to generate various views...
[INFO]: Streaming out GDSII with Magic (log: designs/riscv32/runs/RUN_2026.07.01.16.52.12/logs/signoff/33-gdsii.log)...
[INFO]: Generating MAGLEF views...
[INFO]: Generating lef with Magic (/openlane/designs/riscv32/runs/RUN_2026.07.01.16.52.12/logs/signoff/33-lef.log)...
[STEP 34]
[INFO]: Streaming out GDSII with KLayout (log: designs/riscv32/runs/RUN_2026.07.01.16.52.12/logs/signoff/34-gdsii-klayout.log)...
[STEP 35]
[INFO]: Running XOR on the layouts using KLayout (log: designs/riscv32/runs/RUN_2026.07.01.16.52.12/logs/signoff/35-xor.log)...
[INFO]: No XOR differences between KLayout and Magic gds.
[STEP 36]
[INFO]: Running Magic Spice Export from LEF (log: designs/riscv32/runs/RUN_2026.07.01.16.52.12/logs/signoff/36-spice.log)...
[STEP 37]
[INFO]: Writing Powered Verilog (logs: designs/riscv32/runs/RUN_2026.07.01.16.52.12/logs/signoff/37-write_powered_def.log, designs/riscv32/runs/RUN_2026.07.01.16.52.12/logs/signoff/37-write_powered_verilog.log)...
[STEP 38]
[INFO]: Writing Verilog (log: designs/riscv32/runs/RUN_2026.07.01.16.52.12/logs/signoff/37-write_powered_verilog.log)...
[STEP 39]
[INFO]: Running LVS (log: designs/riscv32/runs/RUN_2026.07.01.16.52.12/logs/signoff/39-lvs.lef.log)...
[STEP 40]
[INFO]: Running Magic DRC (log: designs/riscv32/runs/RUN_2026.07.01.16.52.12/logs/signoff/40-drc.log)...
[INFO]: Converting Magic DRC database to various tool-readable formats...
[INFO]: No DRC violations after GDS streaming out.
[STEP 41]
[INFO]: Running OpenROAD Antenna Rule Checker (log: designs/riscv32/runs/RUN_2026.07.01.16.52.12/logs/signoff/41-arc.log)...
[STEP 42]
[INFO]: Running Circuit Validity Checker ERC (log: designs/riscv32/runs/RUN_2026.07.01.16.52.12/logs/signoff/42-erc_screen.log)...
[INFO]: Saving current set of views in 'designs/riscv32/runs/RUN_2026.07.01.16.52.12/results/final'...
[INFO]: Saving runtime environment...
[INFO]: Generating final set of reports...
[INFO]: Created manufacturability report at 'designs/riscv32/runs/RUN_2026.07.01.16.52.12/reports/manufacturability.rpt'.
[INFO]: Created metrics report at 'designs/riscv32/runs/RUN_2026.07.01.16.52.12/reports/metrics.csv'.
[INFO]: There are no max slew, max fanout or max capacitance violations in the design at the typical corner.
[WARNING]: There are hold violations in the design at the typical corner. Please refer to 'designs/riscv32/runs/RUN_2026.07.01.16.52.12/reports/signoff/31-rcx_sta.min.rpt'.
[INFO]: There are no setup violations in the design at the typical corner.
[SUCCESS]: Flow complete.
[INFO]: Note that the following warnings have been generated:
[WARNING]: Current core area is too small for the power grid settings chosen. The power grid will be scaled down.
[WARNING]: Module sky130_fd_sc_hd__tapvprvrgnd_1 blackboxed during sta
[WARNING]: Module sky130_fd_sc_hd__fill_2 blackboxed during sta
[WARNING]: Module sky130_fd_sc_hd__fill_1 blackboxed during sta
[WARNING]: Module sky130_ef_sc_hd__decap_12 blackboxed during sta
[WARNING]: VSRRC_LOC_FILES is not defined. The IR drop analysis will run, but the values may be inaccurate.
[WARNING]: There are hold violations in the design at the typical corner. Please refer to 'designs/riscv32/runs/RUN_2026.07.01.16.52.12/reports/signoff/31-rcx_sta.min.rpt'.

OpenLane Container (ff5609f):/openlane$ ./flow.tcl -design riscv32
```



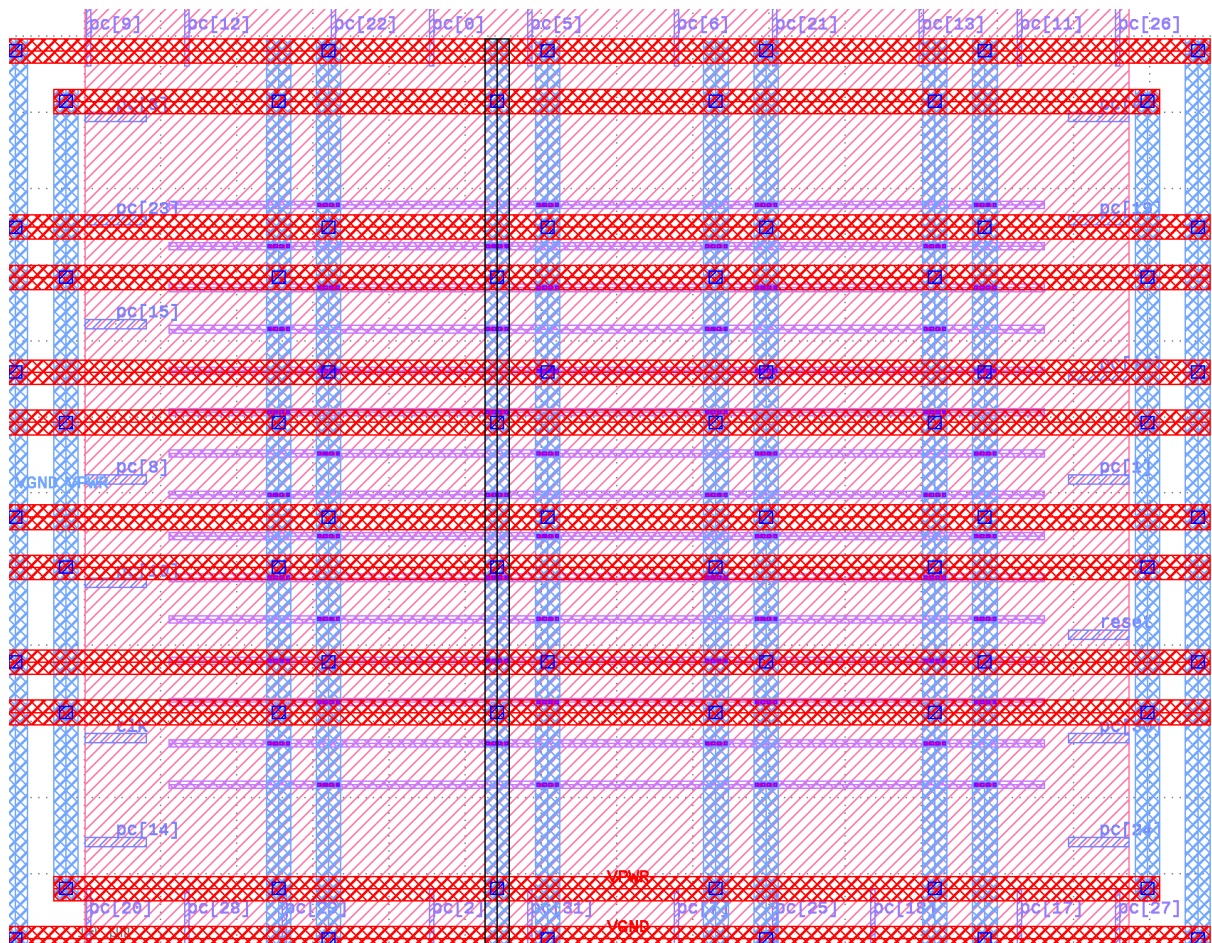
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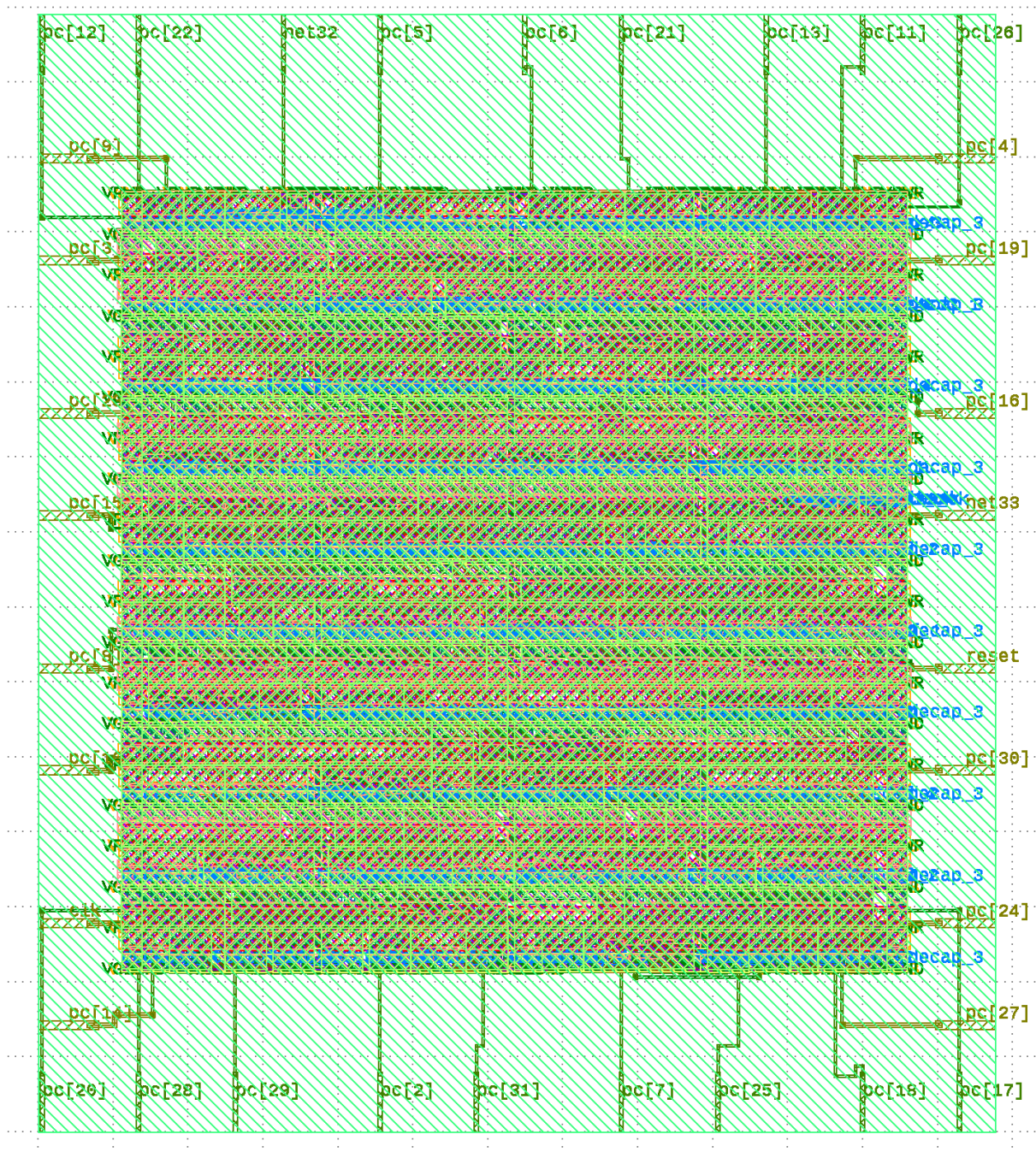






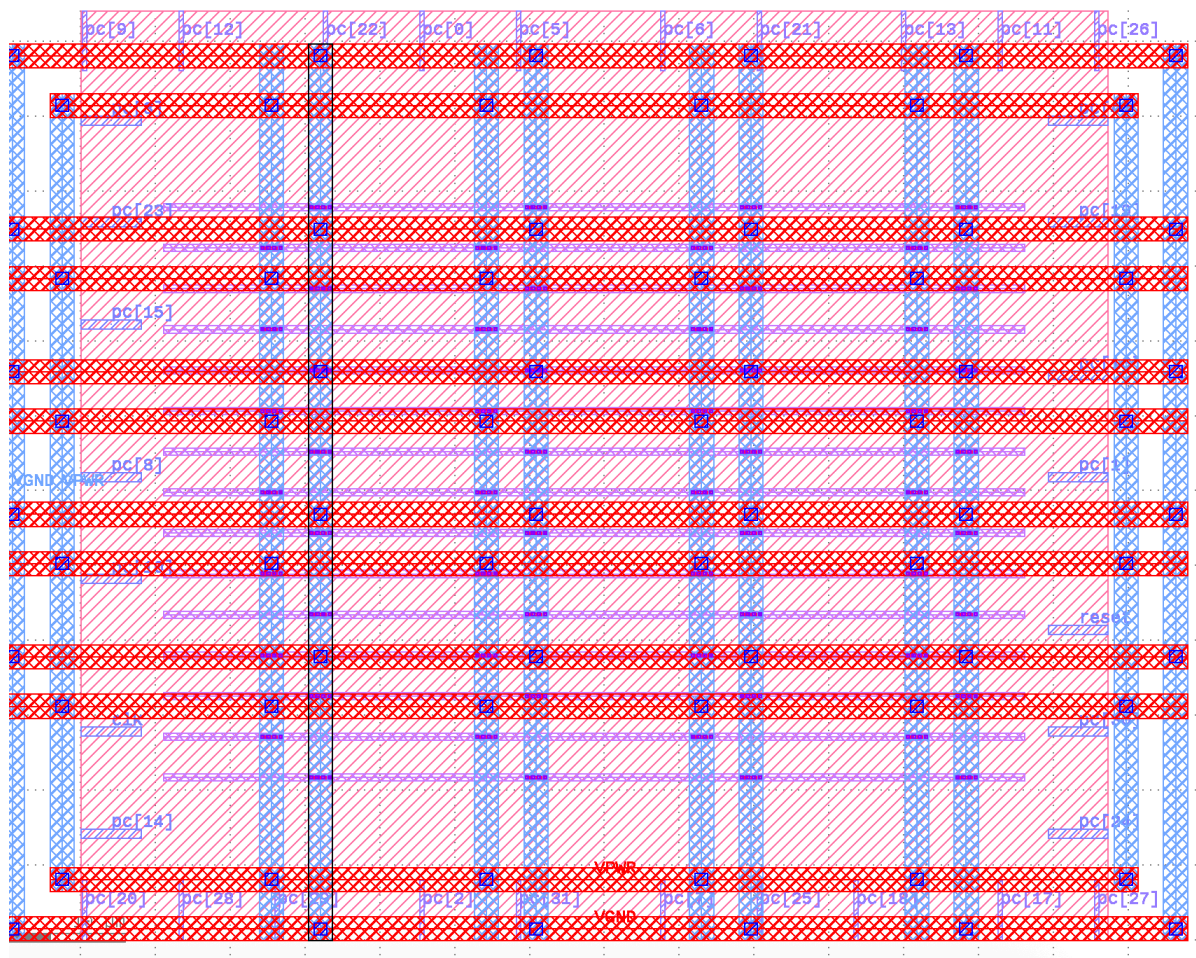
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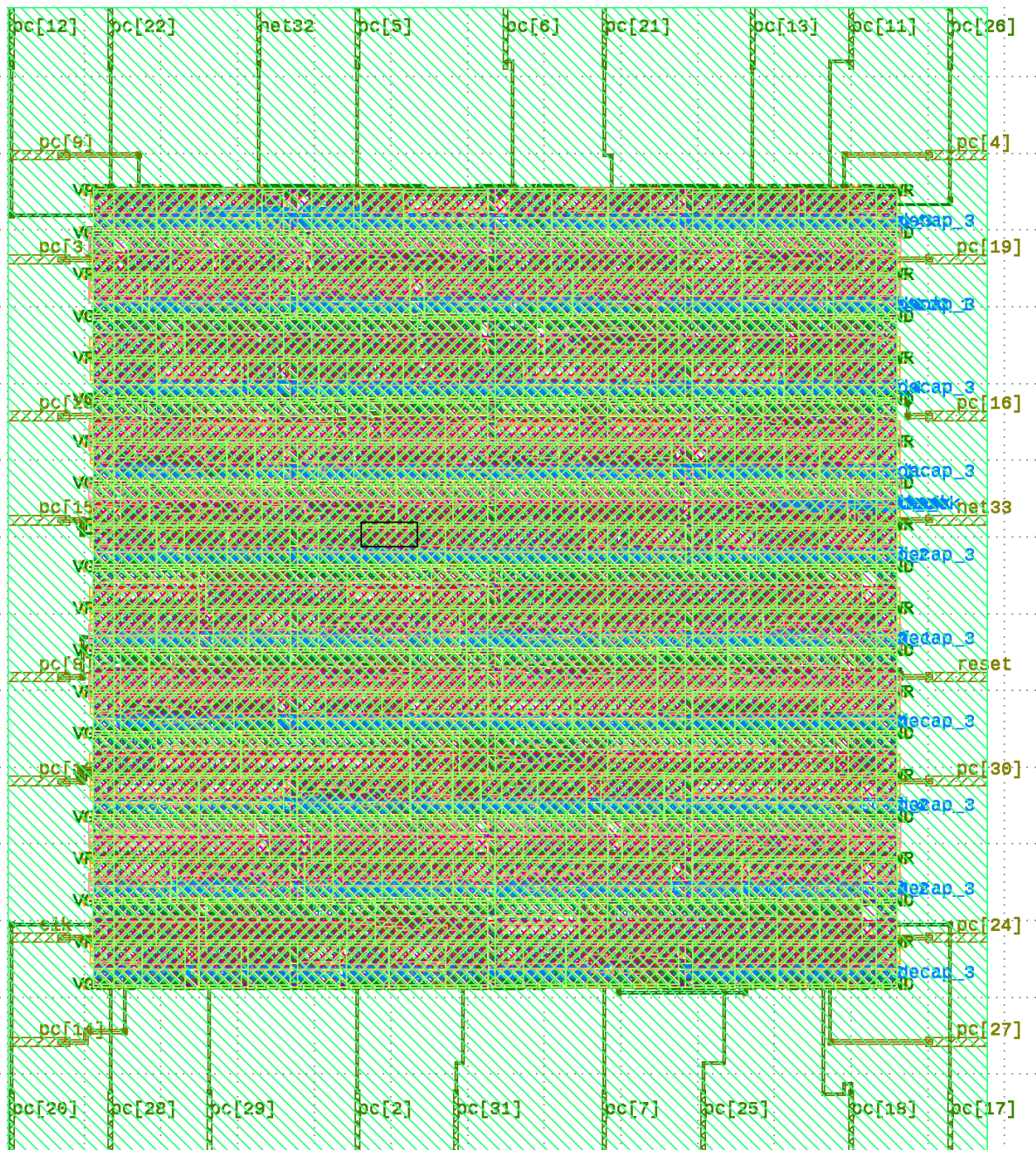






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Requirement

1. Synthesis uncertainty = 15% clock period
CTS uncertainty = 5% clock period
2. Derate = 3%
3. Input/output delay = 50% clock period
4. Utilization = 60%

OpenLane Setting

SYNTH_CLOCK_UNCERTAINTY
 CTS_CLK_UNCERTAINTY
 PL_SET_PE_CORE / OpenSTA derate setting
 SDC constraints
 FP_CORE_UTIL = 60



Requirement

5. Floorplan ratio = 0.7

6. Inputs one side, outputs other side, Metal4

7. Enable core ring

8. CTS target skew = 5% clock period

9. Clock network delay inserted after CTS

OpenLane Setting

FP_ASPECT_RATIO = 0.7

FP_PIN_ORDER_CFG + FP_IO_HLAYER = met4

FP_PDN_CORE_RING = 1

CTS_TARGET_SKEW

CTS_CLK_MAX_WIRE_LENGTH / post CTS settings

results/

— synthesis	→ RTL to gate-level netlist
— floorplan	→ Chip/core planning
— placement	→ Standard cell placement
— cts	→ Clock Tree Synthesis
— routing	→ Metal routing
— signoff	→ Final checks
— final	→ Final ASIC outputs

def	→ Final physical layout description
gds	→ Final chip layout database (fabrication format)
lef	→ Library exchange format
mag	→ Magic layout files
sdc	→ Timing constraints
spef	→ Parasitic extraction data
verilog	→ Final netlist
lib	→ Standard cell libraries
sdf	→ Timing delay data
spi	→ Spice files

2>OpenLane Physical Design Analysis of RISC-V Using KLayout

Aim

To analyze the different stages of ASIC physical design flow (Floorplan, Placement, CTS, and Routing) using OpenLane generated files and visualize the final layout using KLayout.

Tools Used

- OpenLane
 - SKY130 PDK
 - KLayout
 - Docker Environment
-



1. Floorplan Analysis

Objective

To analyze chip dimensions, core area, and physical-only cells.

Input File:

```
results/floorplan/riscv32.def
```

Measurements

Parameter	Value
Die Width	68.62 μm
Die Height	62.065 μm
Core Width	57.5 μm
Core Height	62.56 μm
Physical-only Cells	60

Physical-only Cells Identified

- Tap cells
- Decap cells
- Filler cells
- Endcap cells

These cells improve power integrity, reliability, and manufacturing requirements.

2. Placement Analysis

Objective

To analyze standard cell placement and identify buffers, inverters, and flip-flops.

Input File:

```
results/placement/riscv32.def
```

Cell Analysis



Cell Type Count

Buffers 22

Inverters 0

Flip-Flops 30

Site Row Measurement

Parameter	Value
Site Row Width	57.5 μm
Site Row Height	2.72 μm

Placement arranges standard cells in rows while optimizing timing and area.

3. Clock Tree Synthesis (CTS) Analysis

Objective

To analyze clock tree generation and inserted clock buffers.

Input File:

results/cts/riscv32.def

CTS Cell Analysis

Cell Type	Count
Clock Buffers	30
Total CTS Buffers/Inverters	46

CTS inserts clock buffers to:

- Reduce clock skew
- Improve clock transition
- Balance clock distribution

4. Routing Analysis

Objective

To analyze final signal routing and optimization cells.



Input File:

results/routing/riscv32.def

Routing Analysis

Parameter	Value
Routing Buffers/Inverters	11
Vias	6

Routing connects all placed cells using:

- Metal layers
- Vias
- Interconnect paths

Final Layout Visualization

The final GDSII file:

results/signoff/riscv32.gds

was opened using **KLayout**.

KLayout was used to:

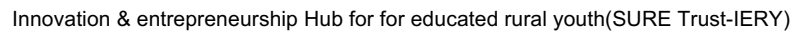
- View complete chip layout
- Zoom into routing details
- Observe different physical layers

Result

The RISC-V design was successfully analyzed through all backend physical design stages. Floorplan dimensions, placement cells, CTS clock buffers, routing elements, and final GDS layout were verified.

Conclusion

OpenLane successfully generated the physical design of the RISC-V processor. KLayout provided a simple method to inspect the final ASIC layout and understand different stages of the VLSI design flow.





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```
mounika@mounikas-MacBook-Air OpenLane % grep -E "tap|decap|fill|filler|endcap" designs/riscv32/runs/run1/results/floorplan/riscv32.def | wc -l
60
mounika@mounikas-MacBook-Air OpenLane % grep -r "sky130_fd_sc_hd_and" /pdk/sky130A/libs.ref/sky130_fd_sc_hd/
grep: /pdk/sky130A/libs.ref/sky130_fd_sc_hd/: No such file or directory
mounika@mounikas-MacBook-Air OpenLane % head -20 designs/riscv32/runs/run1/results/floorplan/riscv32.def
VERSION 5.8 ;
DIVIDERCHAR "/" ;
BUSBITCHARS "{}" ;
DESIGN riscv32 ;
UNITS DISTANCE MICRONS 1000 ;
DIEAREA ( 0 0 ) ( 68620 62065 ) ;
ROW ROW_0 unithd 5520 10880 N DO 125 BY 1 STEP 460 0 ;
ROW ROW_1 unithd 5520 13600 FS DO 125 BY 1 STEP 460 0 ;
ROW ROW_2 unithd 5520 16320 N DO 125 BY 1 STEP 460 0 ;
ROW ROW_3 unithd 5520 19040 FS DO 125 BY 1 STEP 460 0 ;
ROW ROW_4 unithd 5520 21760 N DO 125 BY 1 STEP 460 0 ;
ROW ROW_5 unithd 5520 24480 FS DO 125 BY 1 STEP 460 0 ;
ROW ROW_6 unithd 5520 27200 N DO 125 BY 1 STEP 460 0 ;
ROW ROW_7 unithd 5520 29920 FS DO 125 BY 1 STEP 460 0 ;
ROW ROW_8 unithd 5520 32640 N DO 125 BY 1 STEP 460 0 ;
ROW ROW_9 unithd 5520 35360 FS DO 125 BY 1 STEP 460 0 ;
ROW ROW_10 unithd 5520 38080 N DO 125 BY 1 STEP 460 0 ;
ROW ROW_11 unithd 5520 40800 FS DO 125 BY 1 STEP 460 0 ;
ROW ROW_12 unithd 5520 43520 N DO 125 BY 1 STEP 460 0 ;
ROW ROW_13 unithd 5520 46240 FS DO 125 BY 1 STEP 460 0 ;
mounika@mounikas-MacBook-Air OpenLane % grep -i "DIEAREA" designs/riscv32/runs/run1/results/floorplan/riscv32.def
DIEAREA ( 0 0 ) ( 68620 62065 ) ;
mounika@mounikas-MacBook-Air OpenLane % grep -E "sky130_fd_sc_hd__(buf|inv)" designs/riscv32/runs/run1/results/placement/riscv32.def | wc -l
22
mounika@mounikas-MacBook-Air OpenLane % grep "sky130_fd_sc_hd__buf" designs/riscv32/runs/run1/results/placement/riscv32.def | wc -l
22
mounika@mounikas-MacBook-Air OpenLane % grep "sky130_fd_sc_hd__inv" designs/riscv32/runs/run1/results/placement/riscv32.def | wc -l
0
mounika@mounikas-MacBook-Air OpenLane % grep "sky130_fd_sc_hd__df" designs/riscv32/runs/run1/results/placement/riscv32.def | wc -l
30
mounika@mounikas-MacBook-Air OpenLane % grep -E "sky130_fd_sc_hd__(buf|inv)" designs/riscv32/runs/run1/results/placement/riscv32.def | wc -l
22
mounika@mounikas-MacBook-Air OpenLane % grep "sky130_fd_sc_hd__df" designs/riscv32/runs/run1/results/placement/riscv32.def | wc -l
30
mounika@mounikas-MacBook-Air OpenLane % find ~/OpenLane -name "sky130_fd_sc_hd__buf_1.lef"
mounika@mounikas-MacBook-Air OpenLane %
mounika@mounikas-MacBook-Air OpenLane % head -20 designs/riscv32/runs/run1/results/floorplan/riscv32.def
VERSION 5.8 ;
DIVIDERCHAR "/" ;
BUSBITCHARS "{}" ;
DESIGN riscv32 ;
UNITS DISTANCE MICRONS 1000 ;
DIEAREA ( 0 0 ) ( 68620 62065 ) ;
ROW ROW_0 unithd 5520 10880 N DO 125 BY 1 STEP 460 0 ;
ROW ROW_1 unithd 5520 13600 FS DO 125 BY 1 STEP 460 0 ;
ROW ROW_2 unithd 5520 16320 N DO 125 BY 1 STEP 460 0 ;
ROW ROW_3 unithd 5520 19040 FS DO 125 BY 1 STEP 460 0 ;
ROW ROW_4 unithd 5520 21760 N DO 125 BY 1 STEP 460 0 ;
ROW ROW_5 unithd 5520 24480 FS DO 125 BY 1 STEP 460 0 ;
ROW ROW_6 unithd 5520 27200 N DO 125 BY 1 STEP 460 0 ;
ROW ROW_7 unithd 5520 29920 FS DO 125 BY 1 STEP 460 0 ;
ROW ROW_8 unithd 5520 32640 N DO 125 BY 1 STEP 460 0 ;
ROW ROW_9 unithd 5520 35360 FS DO 125 BY 1 STEP 460 0 ;
ROW ROW_10 unithd 5520 38080 N DO 125 BY 1 STEP 460 0 ;
ROW ROW_11 unithd 5520 40800 FS DO 125 BY 1 STEP 460 0 ;
ROW ROW_12 unithd 5520 43520 N DO 125 BY 1 STEP 460 0 ;
ROW ROW_13 unithd 5520 46240 FS DO 125 BY 1 STEP 460 0 ;
mounika@mounikas-MacBook-Air OpenLane % grep -i "CORE" designs/riscv32/runs/run1/results/floorplan/riscv32.def
mounika@mounikas-MacBook-Air OpenLane % grep -i "ROW" designs/riscv32/runs/run1/results/floorplan/riscv32.def | head
ROW ROW_0 unithd 5520 10880 N DO 125 BY 1 STEP 460 0 ;
ROW ROW_1 unithd 5520 13600 FS DO 125 BY 1 STEP 460 0 ;
ROW ROW_2 unithd 5520 16320 N DO 125 BY 1 STEP 460 0 ;
ROW ROW_3 unithd 5520 19040 FS DO 125 BY 1 STEP 460 0 ;
ROW ROW_4 unithd 5520 21760 N DO 125 BY 1 STEP 460 0 ;
ROW ROW_5 unithd 5520 24480 FS DO 125 BY 1 STEP 460 0 ;
ROW ROW_6 unithd 5520 27200 N DO 125 BY 1 STEP 460 0 ;
ROW ROW_7 unithd 5520 29920 FS DO 125 BY 1 STEP 460 0 ;
ROW ROW_8 unithd 5520 32640 N DO 125 BY 1 STEP 460 0 ;
ROW ROW_9 unithd 5520 35360 FS DO 125 BY 1 STEP 460 0 ;
mounika@mounikas-MacBook-Air OpenLane % grep -E "tap|decap|fill|filler|endcap" designs/riscv32/runs/run1/results/floorplan/riscv32.def | wc -l
60
mounika@mounikas-MacBook-Air OpenLane % find ~/OpenLane -name "sky130_fd_sc_hd__buf_1.lef"
mounika@mounikas-MacBook-Air OpenLane % grep -E "sky130_fd_sc_hd__(buf|inv)" designs/riscv32/runs/run1/results/placement/riscv32.def | wc -l
22
mounika@mounikas-MacBook-Air OpenLane % grep "sky130_fd_sc_hd__buf" designs/riscv32/runs/run1/results/placement/riscv32.def | wc -l
22
mounika@mounikas-MacBook-Air OpenLane % grep "sky130_fd_sc_hd__inv" designs/riscv32/runs/run1/results/placement/riscv32.def | wc -l
0
mounika@mounikas-MacBook-Air OpenLane % grep "sky130_fd_sc_hd__df" designs/riscv32/runs/run1/results/placement/riscv32.def | wc -l
30
mounika@mounikas-MacBook-Air OpenLane % grep "clkbuf" designs/riscv32/runs/run1/results/cts/riscv32.def | wc -l
30
mounika@mounikas-MacBook-Air OpenLane % grep -E "sky130_fd_sc_hd__(clkbuf|buf|inv)" designs/riscv32/runs/run1/results/cts/riscv32.def | wc -l
46
mounika@mounikas-MacBook-Air OpenLane % designs/riscv32/runs/run1/results/cts/riscv32.def
zsh: permission denied: designs/riscv32/runs/run1/results/cts/riscv32.def
mounika@mounikas-MacBook-Air OpenLane % grep "clkbuf" designs/riscv32/runs/run1/results/cts/riscv32.def | wc -l
30
mounika@mounikas-MacBook-Air OpenLane % grep -E "sky130_fd_sc_hd__(clkbuf|buf|inv)" designs/riscv32/runs/run1/results/cts/riscv32.def | wc -l
46
mounika@mounikas-MacBook-Air OpenLane % grep -E "sky130_fd_sc_hd__(buf|inv)" designs/riscv32/runs/run1/results/routing/riscv32.def | wc -l
11
mounika@mounikas-MacBook-Air OpenLane % grep "VIA" designs/riscv32/runs/run1/results/routing/riscv32.def | wc -l
6
mounika@mounikas-MacBook-Air OpenLane % grep "^ - " designs/riscv32/runs/run1/results/routing/riscv32.def | wc -l
0
mounika@mounikas-MacBook-Air OpenLane %
```